

2a) b) $\vec{x}_k = (\vec{x}_k \cdot \vec{v}_i) \vec{v}_i + \vec{r}_k$

Converting vector k to eigen vector coordinate system

taking the dot product with itself

we get

$$\vec{r}_k \cdot \vec{r}_k = \left(\sum_i a_i \lambda_i^k \vec{v}_i \right) \cdot \left(\sum_j a_j \lambda_j^k \vec{v}_j \right) = \sum_i \sum_j (a_i \lambda_i^k) (a_j \lambda_j^k) (\vec{v}_i \cdot \vec{v}_j)$$

since $a_i \cdot a_j = 0$ (perpendicular)

$$a_i \cdot a_i = 1$$

$$|\vec{r}_k|^2 = \sum_i a_i^2 \lambda_i^{2k}$$

$$\vec{r}_k = \sqrt{\sum_i a_i^2 \lambda_i^{2k}}$$

$$\rho_k = \frac{\sqrt{\sum_i a_i^2 \lambda_i^{2k}}}{|a| |\lambda|^{2k}}$$

$$\rho_{k+1} = \frac{\sqrt{\sum_i a_i^2 \lambda_i^{2k+2}}}{|a| |\lambda|^{2k+2}}$$

taking ratio $\frac{\rho_{k+1}}{\rho_k}$

$$\frac{\sqrt{\sum_i a_i^2 \lambda_i^{2k+2}}}{|a| |\lambda|^{2k+2}} \times \frac{|a| |\lambda|^{2k}}{\sqrt{\sum_i a_i^2 \lambda_i^{2k}}}$$

$$= \frac{1}{|\lambda|} \sqrt{\frac{\sum_i a_i^2 \lambda_i^{2k+2}}{\sum_i a_i^2 \lambda_i^{2k}}}$$

$$\text{let } a_i^2 \lambda_i^{2k} = \omega_i$$

$$= \frac{1}{|\lambda|} \sqrt{\sum_i \omega_i \times \lambda_i^2}$$

This is the form of average
always $\leq$ the greatest value.

λ_2 is greatest value

We have to prove that $\vec{r}_k$ tends to 0
so that $\vec{x}_k \rightarrow \vec{v}_1$

$$\text{let } \rho_k = \frac{|\vec{r}_k|}{|\vec{x}_k \cdot \vec{v}_1|}$$

$$\text{if } \rho_k \rightarrow 0, \vec{x}_k \rightarrow \vec{v}_1$$

We do not have to normalize the vector since we are taking the ratio, it doesn't change

So let $\vec{y}_k$ be the iterator

$$\vec{y}_k = X \times X \times \dots \times \vec{x}_0 = X^k \vec{x}_0$$

Converting x_0 to eigebasis

allows us to write

$$\vec{x}_0 \text{ as } \vec{x}_0 = \sum_i a_i \vec{v}_i$$

$$X \cdot \vec{x}_0 = \sum_i a_i X \vec{v}_i = \sum_i a_i \lambda_i \vec{v}_i$$

for x^k

$$\vec{y}_k = X^k \vec{x}_0 = \sum_i a_i \lambda_i^k \vec{v}_i$$

In $\vec{y}_k$, the $i=1$ term is the first term $(a_1 \cdot \lambda_1) \vec{v}_1$

everything else is the residual

$$\vec{r}_k = \sum_i a_i \lambda_i^k \vec{v}_i = a_2 \lambda_2^k \vec{v}_2 + a_3 \lambda_3^k \vec{v}_3 + \dots + a_n \lambda_n^k \vec{v}_n$$

$$\therefore \frac{\sum w_i \lambda_i^2}{\sum w_i} \leq \lambda_2^2$$

$$\frac{p_{k+1}}{p_k} = \frac{1}{|\lambda_1|} \sqrt{\dots} \leq \frac{1}{|\lambda_1|} \sqrt{(\lambda_2)^2} \leq \boxed{\frac{|\lambda_2|}{|\lambda_1|}}$$

$$\frac{p_{k+1}}{p_k} \leq \frac{|\lambda_2|}{|\lambda_1|}, \text{ therefore each iteration shrinks by } \frac{|\lambda_2|}{|\lambda_1|}$$

↗

which implies $\rightarrow p_k \rightarrow 0$

$$r_k \rightarrow 0$$

$$\hookrightarrow x_k = \pm v_k$$

$$2c) i) X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\vec{x}_0 = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\vec{x}_1 = X \vec{x}_0 = \begin{bmatrix} a \\ -b \end{bmatrix}$$

$$\vec{x}_2 = X \vec{x}_1 = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\vec{x}_3 = X \vec{x}_2 = \begin{bmatrix} a \\ -b \end{bmatrix}$$

so for an arbitrary value of k

$\vec{x}_k$ oscillates between $\begin{bmatrix} a \\ b \end{bmatrix}$ and $\begin{bmatrix} a \\ -b \end{bmatrix}$

$$\vec{x}_k = \begin{bmatrix} a \\ (-1)^k b \end{bmatrix}$$

$\therefore \vec{x}_k$ never converges to a single vector
it oscillates between $\begin{bmatrix} a \\ b \end{bmatrix}$ and $\begin{bmatrix} a \\ -b \end{bmatrix}$

ii) iii) Explanation

2d)

i) let X be $m \times n$ matrix

such that

$$\begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}$$

X^T will be a $n \times m$ matrix

$X \cdot X^T$ will be a $m \times m$ matrix
which is a symmetric matrix

similarly $X^T \cdot X$ will be a $n \times n$ matrix
which is a symmetric matrix

let $\vec{w}$ be every real vector

if $[\vec{w}^T \cdot A \vec{w}] \geq 0$, then A is a PSD

$X \cdot X^T$ is an m dimensional square matrix

let $\vec{w}$ be an m dimensional vector

$$\vec{w}^T \cdot X^T X \vec{w}$$

$$= (\vec{w} X)^T \cdot (\vec{w} X) = \|X^T \vec{w}\|^2 \geq 0$$

ii) Suppose $\vec{v}$ is an eigenvector of $X^T X$ with $\lambda \neq 0$

$$X^T X \vec{v} = \lambda \vec{v}, \lambda \neq 0$$

$$X (X^T X \vec{v}) = X \lambda \vec{v}$$

$$X X^T \cdot X \vec{v} = \lambda X \vec{v}$$

$X \vec{v}$ is eigenvector of $X X^T$
with eigenvalue λ

Suppose μ is an eigenvalue of $X X^T$ and $\mu \neq 0$

$$X X^T \vec{u} = \mu \vec{u}, \mu \neq 0$$

$$X^T X \cdot X^T \vec{u} = \mu X^T \vec{u}$$

therefore $X^T \vec{u}$ is an eigenvector
of the $m \times m$ $X^T X$ with eigenvalue μ

since $X \vec{v}$ is an eigenvector of both

$X X^T$ and $X^T X$ therefore $\lambda = \mu$

and therefore they have the same
eigenvalues $(\sigma)^2$

2) e)

Let function $E(X)$ take an arbitrary
non symmetrical matrix X with $n \times m$ and return
 $\sigma_1, \vec{v}_1, \vec{u}_1$.

if $n \gg m$ compute $X X^T$

else if $m \gg n$ compute $X^T X$

Reason: $X X^T$ has dimensions $m \times m$
and since $m \ll n$
cost less computationally

similarly ...

Applying power method to $XX^T \dots$

take an arbitrary unit vector pointing random direction

Using the XX^T matrix as a transformation repeatedly apply it to the arbitrary initial vector until it converges with tolerance value or a set number of iterations. Remember to normalize the vector.

After ... $\vec{v}_1$ is an eigenvector.

We know

$$XX^T \cdot \vec{v}_1 = \lambda_1 \cdot \vec{v}_1$$

$$\det(XX^T - \lambda I) = 0$$

solve for λ

$$\sqrt{\lambda} = \sigma$$

$$\vec{u}_1 = \frac{X \vec{v}_1}{\sigma_1}$$

return values of $\vec{u}_1, \vec{v}_1, \sigma_1$

ii) $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ 3×2 matrix

We need to compute $X^T \cdot X$ which gives a 2×2 matrix

$$X^T \cdot X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

take on arbitrary vector $\vec{x}$

$$X^T X \vec{x} = \lambda \vec{x}$$

$$\det(X^T X - \lambda I) = 0$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$\det = (2-\lambda)^2 - 1 = 0$$

$$4 + \lambda^2 - 4\lambda - 1 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow \lambda(\lambda - 3) - 1(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 3) = 0$$

Let $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ (or) $\lambda = 3$

from $(M - \lambda I) \vec{v} = 0,$

$$\begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= 0$$

$$\Rightarrow (2 - \lambda)x + y = 0$$

$$x + (2 - \lambda)y = 0$$

$$(2 - \lambda)^2 y - y = 0$$

$$y((2 - \lambda)^2 - 1) = 0$$

$\therefore$ dominant λ is 3,

$$M = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = y, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \vec{v}_1$$

$\sqrt{2}$ is length of vector $\vec{v}_1$

$$\sigma_1 = \sqrt{3}$$

$$\vec{v}_1' = \frac{\vec{v}_1}{\sigma_1}$$

$$= \frac{1}{\sqrt{3}} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\sigma_1 \overline{U}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$= X \times \overline{U}_1$$

[similarly computing for $X \times X^T$
if $N < M$]

$$X^T \overline{U}_1 = \sigma_1 \overline{V}_1$$